Shakti Kumar

Computational Materials Scientist · AI Training Task Author · PhD Scholar, IIT Kanpur

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Profile

PhD scholar in Materials Science & Engineering at IIT Kanpur (creep and high-temperature deformation of alloys), with an M.Tech in Aerospace and a B.Tech in Mechanical Engineering. Builds reproducible computational-materials workflows in Python (DFT, MD, phase-field, ML), each unit-tested and checked against experiment or literature. Authors AI training tasks with deterministic Python graders; uses AI coding tools openly and verifies their output against physics and published data.

Experience

AI Training Task Author (freelance) — Alignerr (Labelbox) · 2026 – present

- Authoring an engineering reinforcement-learning task for frontier AI models: problem design, reference solution and a deterministic Python grader, reviewed and tested through a Git/CI workflow.

Teaching: Graduate Teaching Assistant, IIT Kanpur (Jul 2023 – May 2024; Dec 2025 – present), tutorials and rubric-based evaluation · Teaching Faculty, GeeksforGurukul, Indore (Nov 2024 – May 2025) · Unacademy (2021–22), 200+ step-by-step GATE Mechanical solutions.

Computational projects

Computational Materials Science Portfolio — github.com/Shaktikr/mse-computational-projects · 2026 · 11 projects, 60 unit tests

- Computed aluminium's creep activation energy from first principles in Quantum ESPRESSO (141 converged runs): vacancy formation 0.67 eV plus migration 0.59 eV gives Q = 1.26 eV, against 1.28 eV measured.
- Reproduced Ogata, Li & Yip (Science, 2002) with DFT: aluminium is stronger than copper in ideal shear (3.19 vs 2.14 GPa).

- Calculated B2 NiAl elastic constants within 5% of experiment and ran seed-averaged LAMMPS creep simulations of nanocrystalline Ni.
- Reached 80% phase-prediction accuracy on a real 1,545-entry high-entropy-alloy dataset, against 58% for empirical rules.

ML Stress Prediction for Gyroid TPMS Lattices — github.com/Shaktikr/gyroid-fea-ml · MSE643 course project, IIT Kanpur

- Built a reproducible pipeline comparing five regressors on Abaqus compression data (5 geometries, 2,525 records); a feature-engineered linear model beat tree ensembles in the elastic regime (test MAE 0.14 MPa).

Experimental research

Thermal-Shock Testing of Alumina TBCs — github.com/Shaktikr/ceramic-coating-thermal-fatigue · B.Tech project, team of five, 2021–22

- Tested eight plasma-sprayed alumina compositions above 1100 °C: only 20 wt% 3YSZ survived, and survival tracked cooling rate (2.10 vs 4.25 °C/s).

M.Tech Thesis: Compressible Elliptic Vortex Ring Dynamics — IIT Kanpur, 2022–24: shock-tube experiments with high-speed schlieren imaging; image-based flow analysis in MATLAB and Python.

Education

- **PhD, Materials Science & Engineering (ongoing) — IIT Kanpur** · Dec 2025 – present. Coursework: Thermodynamics of Materials; Fracture, Fatigue and Creep; AI & ML in Materials Engineering; Mathematics & Computational Methods; ongoing: Transport Phenomena, Microscopy & Microanalysis.
- **M.Tech, Aerospace Engineering — IIT Kanpur** · 2022 – 2024
- **B.Tech, Mechanical Engineering — REC Banda** · 2018 – 2022 · CPI 7.85
- **GATE 2021 (Mechanical)**: AIR 8,664 of 1,20,594 (top 7.2%); GATE 2024 (Computer Science): AIR 22,651

Skills

- **Simulation:** Quantum ESPRESSO, ASE, pymatgen, LAMMPS, phase-field modelling, ANSYS; command-line workflows; structure–property, creep and fatigue analysis
- **Programming:** Python (NumPy, SciPy, pandas, scikit-learn, matplotlib, pytest), MATLAB, C/C++, SQL, Git, LaTeX